

Detector Coherence Transfer in Quantum Measurement

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March 2026

Abstract

We present a detector-centered framework for quantum measurement using QMCTB-01 and QMCTB-02 benchmarks. Measured values are $V(\sigma=0)=0.966 \pm 0.002$, $V(\sigma=\pi)=0.189 \pm 0.032$, $\sigma_c=2.339 \pm 0.009$, and $R^2=0.9990$. Interference visibility is governed by detector-plane coherence preservation.

1. Introduction

Detector architecture determines observed interference visibility, not accumulation.

2. Theoretical Framework

Results arise from detector-plane signal processing, independent of interpretation.

3. Methods

Gaussian phase noise applied: $\phi \sim N(0, \sigma^2)$. TEM00 Gaussian beam used (fixed encoding).

4. Results

V_{high} (fit) differs slightly from measured $V(\sigma=0)$ due to finite sampling.

5. Discussion

Detector-plane acts as coherence filter defining measurement outcome.

6. Unified Framework

Field $\rightarrow$ imprint $\rightarrow$ detector plane $\rightarrow$ transfer function $\rightarrow$ measurement.

7. Conclusion

Detector coherence is a measurable hardware parameter controlling visibility.